INTERNSHIP PROJECT REPORT

ON

Hotel reservation system using KNN classifier

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This project has enhanced our practical understanding of machine learning and strengthened our problem-solving skills, and we are grateful for this opportunity.

ABSTRACT

The rapid growth of online hotel booking platforms has led to a significant increase in reservation cancellations, resulting in revenue loss and inefficient resource management for hotels. This project focuses on predicting hotel booking cancellations using a machine learning approach, specifically the K-Nearest Neighbors (KNN) algorithm.

The system utilizes historical booking data, including features such as lead time, room type, pricing, and customer details, to identify patterns associated with cancellations. Data preprocessing techniques such as handling missing values, encoding categorical variables, and feature scaling are applied to prepare the dataset for modeling. Additionally, class imbalance is addressed using SMOTE to improve prediction performance.

The KNN model is trained and evaluated using standard metrics such as accuracy, precision, recall, and F1-score. The results demonstrate that the model can effectively classify bookings as canceled or not canceled, providing valuable insights for hotel management.

This predictive system can help hotels optimize pricing strategies, reduce revenue loss, and improve operational efficiency. The project highlights the importance of machine learning in solving real-world business problems in the hospitality industry.

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INTRODUCTION

The hospitality industry has undergone a significant transformation with the rise of online booking platforms and digital reservation systems. While these advancements have made hotel bookings more convenient for customers, they have also introduced challenges, particularly in the form of high cancellation rates. Booking cancellations directly impact hotel revenue, disrupt operational planning, and lead to inefficient utilization of resources such as rooms, staff, and inventory.

In recent years, hotels have increasingly relied on data-driven approaches to address these challenges. Machine learning, a key domain of data science, offers powerful tools to analyze historical data, identify hidden patterns, and make accurate predictions. This project, titled *"Hotel Reservation Cancellation Prediction using K-Nearest Neighbors (KNN) Classifier,"* focuses on leveraging machine learning techniques to predict whether a hotel booking will be canceled or not.

The primary goal of this project is to build a predictive model that can assist hotel management in making informed decisions. By analyzing factors such as booking lead time, customer type, pricing, room preferences, and market segment, the model aims to classify reservations into two categories: canceled and not canceled. Early prediction of cancellations allows hotels to implement strategies such as dynamic pricing, controlled overbooking, and targeted customer engagement to minimize losses.

The K-Nearest Neighbors (KNN) algorithm is chosen for this project due to its simplicity, effectiveness, and intuitive approach to classification. KNN is a non-parametric, instance-based learning algorithm that classifies data points based on the similarity to their nearest neighbors. Despite its simplicity, it performs well in many real-world classification problems when combined with proper preprocessing and feature scaling.

The project involves a complete machine learning pipeline, including data collection, data preprocessing, exploratory data analysis (EDA), feature engineering, model training, and evaluation. Special attention is given to handling imbalanced data using techniques like SMOTE, as imbalance can significantly affect model performance.

Through this project, we aim to demonstrate how machine learning can be effectively applied in the hospitality sector to solve practical business problems. The outcome of this system can support hotel management in improving operational efficiency, enhancing customer satisfaction, and maximizing revenue.

Overall, this project not only provides a predictive solution but also offers valuable insights into customer booking behavior, highlighting the growing importance of data analytics in modern business environments.

PROBLEM STATEMENT

In the hospitality industry, hotel booking cancellations have become a major challenge due to the widespread use of online reservation systems. Customers can easily book and cancel rooms without significant penalties, leading to a high rate of cancellations. This results in revenue loss, inefficient room utilization, and difficulties in demand forecasting for hotel management.

The core problem addressed in this project is to develop a system that can accurately predict whether a hotel reservation will be canceled or not based on historical booking data. The prediction is formulated as a binary classification problem, where each booking is classified as either *canceled* or *not canceled*.

To solve this problem, various factors such as lead time, room type, pricing, customer type, and booking patterns are analyzed using machine learning techniques. By identifying patterns and trends in past data, the system aims to assist hotel management in making proactive decisions, such as optimizing pricing strategies, managing overbooking, and improving resource allocation.

This project specifically focuses on implementing the K-Nearest Neighbors (KNN) algorithm to build an effective and reliable prediction model. The ultimate goal is to minimize the negative impact of cancellations and enhance overall operational efficiency in the hotel industry.

OBJECTIVES

The main objective of this project is to develop a machine learning-based system that can accurately predict hotel booking cancellations using the K-Nearest Neighbors (KNN) algorithm. The detailed objectives of the project are as follows.

The specific objectives of this project are as follows:

1. Data Understanding and Analysis

- To study and understand the structure of the hotel booking dataset
- To identify key features influencing booking cancellations
- To analyze patterns and trends in customer behavior

2. Data Preprocessing

- To handle missing or inconsistent data effectively
- To convert categorical data into numerical format using encoding techniques
- To remove irrelevant or redundant features to improve model performance

3. Exploratory Data Analysis (EDA)

- To visualize relationships between different variables
- To identify important factors such as lead time, pricing, and customer type
- To detect outliers and anomalies in the dataset

4. Feature Engineering and Scaling

- To transform features into a suitable format for machine learning models
- To apply feature scaling techniques (Standardization/Normalization)
- To ensure all features contribute equally to distance-based algorithms like KNN

5. Handling Imbalanced Data

- To address class imbalance in booking status (canceled vs not canceled)
- To apply techniques like SMOTE for better model generalization
- To improve prediction performance for minority class

6. Model Development using KNN

- To implement the K-Nearest Neighbors algorithm
- To select the optimal value of K for better accuracy

- To train the model using processed data

7. Model Evaluation

- To evaluate the model using performance metrics such as:
    - Accuracy
    - Precision
    - Recall
    - F1-score
- To analyze strengths and weaknesses of the model

8. Prediction System

- To build a system that can predict cancellation for new bookings
- To ensure the model can be used for real-world applications

9. Performance Improvement

- To compare results with different parameter values
- To improve model performance through tuning and preprocessing

10. Practical Application

- To help hotel management reduce revenue loss
- To support better decision-making in pricing and booking strategies
- To enhance operational efficiency using predictive analytics

Overall, the project aims to bridge the gap between theoretical machine learning concepts and real-world business applications in the hospitality industry.

DATASET DESCRIPTION

The dataset used in this project consists of historical hotel reservation records, which are utilized to analyze booking patterns and predict whether a reservation will be canceled or not. It contains a mix of numerical and categorical features that describe customer behavior, booking details, and hotel-related information.

1. Overview of Dataset

- The dataset includes multiple records, where each row represents a single hotel booking.
- It contains both **independent variables (features)** and a **dependent variable (target)**.
- The target variable indicates the booking status (canceled or not canceled).
- The dataset is suitable for a **binary classification problem**.

2. Features Description

Below is a detailed explanation of the key attributes used in the dataset:

a) Customer-Related Features

- **no_of_adults**: Number of adults included in the booking
- **no_of_children**: Number of children included in the booking
- **repeated_guest**: Indicates whether the customer has booked before (1 = Yes, 0 = No)

These features help in understanding customer type and loyalty behavior.

b) Booking Details

- **no_of_weekend_nights**: Number of weekend nights booked
- **no_of_week_nights**: Number of weekday nights booked
- **lead_time**: Number of days between booking date and arrival date

Lead time is a crucial feature, as longer lead times are often associated with higher cancellation probability.

c) Room and Service Information

- **type_of_meal_plan**: Type of meal plan selected (e.g., Meal Plan 1, Meal Plan 2)
- **required_car_parking_space**: Whether parking is required (0 or 1)
- **room_type_reserved**: Type/category of room booked

These features reflect customer preferences and service requirements.

d) Booking Channel and Market Information

- **market_segment_type**: Source of booking (Online, Offline, Corporate, etc.)

Different booking channels show different cancellation behaviors, with online bookings often having higher cancellation rates.

e) Pricing Information

- **avg_price_per_room**: Average price of the room per night

Pricing plays a significant role in cancellations, as higher prices may increase the likelihood of cancellation.

f) Special Requests

- **no_of_special_requests**: Number of special requests made by the customer

Customers with more special requests are generally more committed and less likely to cancel.

g) Date-Related Features

- **arrival_year**, **arrival_month**, **arrival_date**

These features help identify seasonal trends and demand patterns.

3. Target Variable

- **booking_status**:
    - **0 → Not Canceled**
    - **1 → Canceled**

This is the output variable that the model aims to predict.

4. Data Types

- **Numerical Features**: Lead time, number of adults, price, number of nights, etc.
- **Categorical Features**: Meal plan, room type, market segment, etc.

Categorical features are converted into numerical form using encoding techniques for model compatibility.

5. Data Quality and Challenges

- Presence of missing or inconsistent values
- Imbalanced dataset (more non-canceled bookings than canceled ones)
- Need for feature scaling due to distance-based algorithm (KNN)

6. Data Preprocessing Requirements

To prepare the dataset for modeling, the following steps are required:

- Handling missing values
- Encoding categorical variables
- Feature scaling (Standardization/Normalization)
- Removing outliers (if necessary)
- Balancing dataset using SMOTE

7. Importance of Dataset

The dataset plays a crucial role in determining model performance. Proper understanding and preprocessing of data help in:

- Improving prediction accuracy
- Reducing model bias
- Enhancing generalization capability

8. Summary

The dataset provides comprehensive information about hotel bookings, including customer demographics, booking behavior, pricing, and service preferences. These features collectively enable the machine learning model to identify patterns and predict booking cancellations effectively.

Overall, the dataset forms the foundation of this project and is essential for building a reliable and accurate predictive system.

Dataset:-

8aa90ba1-0a54-40a3-9248-3ef8333

Booking_ID	no_of_adults	no_of_children	no_of_weekend_nights	no_of_week_nights
INN00001	2	0	1	2
INN00002	2	0	2	3
INN00003	1	0	2	1
INN00004	2	0	0	2
INN00005	2	0	1	1
INN00006	2	0	0	2
INN00007	2	0	1	3
INN00008	2	0	1	3
INN00009	3	0	0	4
INN00010	2	0	0	5
INN00011	1	0	1	0
INN00012	1	0	2	1
INN00013	2	0	2	1
INN00014	1	0	2	0
INN00015	2	0	0	2
INN00016	2	0	0	2
INN00017	1	0	1	0
INN00018	2	0	1	3
INN00019	2	0	2	2
INN00020	2	0	1	0
INN00021	2	0	2	2
INN00022	1	0	0	1
INN00023	2	0	2	4
INN00024	2	0	0	3
INN00025	2	0	2	1
INN00026	2	0	0	2
INN00027	2	0	1	1
INN00028	1	0	0	2
INN00029	1	0	1	2
INN00030	2	0	0	2
INN00031	2	0	1	1
INN00032	2	0	2	2
INN00033	0	2	0	3
INN00034	2	0	0	1
INN00035	2	0	0	2
INN00036	2	0	0	1
INN00037	1	0	2	1
INN00038	2	0	0	1
INN00039	2	0	2	3
INN00040	2	0	2	1
INN00041	2	0	0	4
INN00042	2	0	0	2
INN00043	3	0	2	1
INN00044	2	0	1	1
INN00045	2	0	0	4
INN00046	2	0	0	3
INN00047	1	0	0	2
INN00048	2	0	0	1
INN00049	2	0	0	4
INN00050	1	0	1	3
INN00051	1	0	2	5
INN00052	2	0	2	2
INN00053	1	0	1	2
INN00054	2	0	0	4

METHODOLOGY

The methodology of this project follows a structured and systematic machine learning pipeline designed to transform raw hotel booking data into a predictive model capable of accurately classifying booking cancellations. Each stage is carefully implemented to ensure data quality, model efficiency, and reliable predictions.

1. Data Collection

The dataset is obtained from a reliable source containing historical hotel booking records. Each record represents a booking instance with multiple attributes such as customer details, booking information, and pricing.

- Data is imported using Python libraries such as **Pandas**
- Initial inspection is performed using:
    - .head()
    - .info()
    - .describe()

This step helps in understanding the structure, size, and data types.

2. Data Understanding

Before processing, the dataset is analyzed to identify:

- Number of features and observations
- Data types (numerical/categorical)
- Target variable distribution
- Presence of missing values
- Class imbalance

This stage ensures clarity on how the data should be handled.

3. Data Preprocessing

Data preprocessing is one of the most critical stages, as raw data is often noisy and inconsistent.

3.1 Handling Missing Values

- Missing values are identified using:
  - .isnull().sum()

Treatment methods:

- Numerical columns → replaced with mean/median
- Categorical columns → replaced with mode
- In some cases, rows with excessive missing values are removed

3.2 Removing Duplicates

- Duplicate records are detected using .duplicated()
- Removed to avoid bias in model training

3.3 Encoding Categorical Variables

Machine learning models require numerical input, so categorical features are converted:

- **Label Encoding** → for ordinal categories
- **One-Hot Encoding** → for nominal categories

Example:

- Room Type → A, B, C → [1,0,0], [0,1,0], etc.

3.4 Feature Selection

Irrelevant or redundant features are removed to:

- Reduce dimensionality
- Improve model performance
- Avoid overfitting

Techniques:

- Correlation analysis
- Domain knowledge

4. Exploratory Data Analysis (EDA)

EDA is performed to extract insights and understand relationships between variables.

4.1 Univariate Analysis

- Distribution of individual features
- Histograms and count plots

4.2 Bivariate Analysis

- Relationship between features and target variable

Examples:

- Lead time vs cancellation
- Price vs cancellation

4.3 Correlation Analysis

- Heatmaps used to identify relationships between numerical variables
- Helps in feature selection

Key Insights:

- Higher lead time → higher cancellation probability
- Online bookings → more cancellations
- Repeated customers → lower cancellation rate

5. Feature Scaling

Since KNN is a **distance-based algorithm**, feature scaling is essential.

5.1 Why Scaling is Required

Without scaling:

- Features with larger values dominate distance calculation

5.2 Techniques Used

- **StandardScaler**
  - Transforms data to mean = 0 and variance = 1
- **MinMaxScaler**
  - Scales data between 0 and 1

6. Handling Imbalanced Dataset

The dataset often has:

- More non-canceled bookings
- Fewer canceled bookings

This creates bias in the model.

6.1 SMOTE (Synthetic Minority Oversampling Technique)

- Generates synthetic samples for minority class
- Balances dataset

Steps:

1. Identify minority class
2. Generate synthetic data points
3. Combine with original dataset

6.2 Benefits

- Improves recall
- Reduces bias
- Enhances model generalization

7. Train-Test Split

Dataset is divided into:

- **Training Set (80%)** → Used to train model
- **Testing Set (20%)** → Used to evaluate performance

This ensures unbiased evaluation.

8. Model Building – KNN Classifier

8.1 Algorithm Overview

KNN is a **lazy learning algorithm** that stores all training data and makes predictions based on similarity.

8.2 Working Steps

1. Choose value of K
2. Compute distance between test point and all training points
3. Select K nearest neighbors
4. Perform majority voting

8.3 Distance Metric

- Euclidean distance is used:

  [
  d = \sqrt{\sum (x_i - y_i)^2}
  ]

8.4 Choosing Optimal K

- Small K → Overfitting
- Large K → Underfitting

  Optimal K is selected using:

- Cross-validation
- Accuracy comparison

9. Model Training

- KNN model is trained using scaled training data
- Multiple K values are tested

- Best-performing model is selected

10. Model Evaluation

The trained model is evaluated using multiple metrics:

10.1 Confusion Matrix

	Predicted No	**Predicted Yes**
Actual No	TN	FP
Actual Yes	FN	TP

10.2 Performance Metrics

- Accuracy
- Precision
- Recall
- F1-score

10.3 Interpretation

- High accuracy → good overall performance
- High recall → fewer missed cancellations
- High precision → fewer false alarms

11. Model Optimization

To improve performance:

- Tune K value
- Adjust scaling methods
- Improve feature selection
- Use cross-validation

12. Prediction System

A prediction pipeline is created:

Steps:

1. Input new booking data
2. Apply preprocessing (encoding, scaling)
3. Pass data to trained model
4. Output prediction (0 or 1)

13. Deployment (Optional Extension)

The model can be deployed using:

- Flask / Django (Web App)
- Streamlit (Dashboard)

This allows real-time prediction.

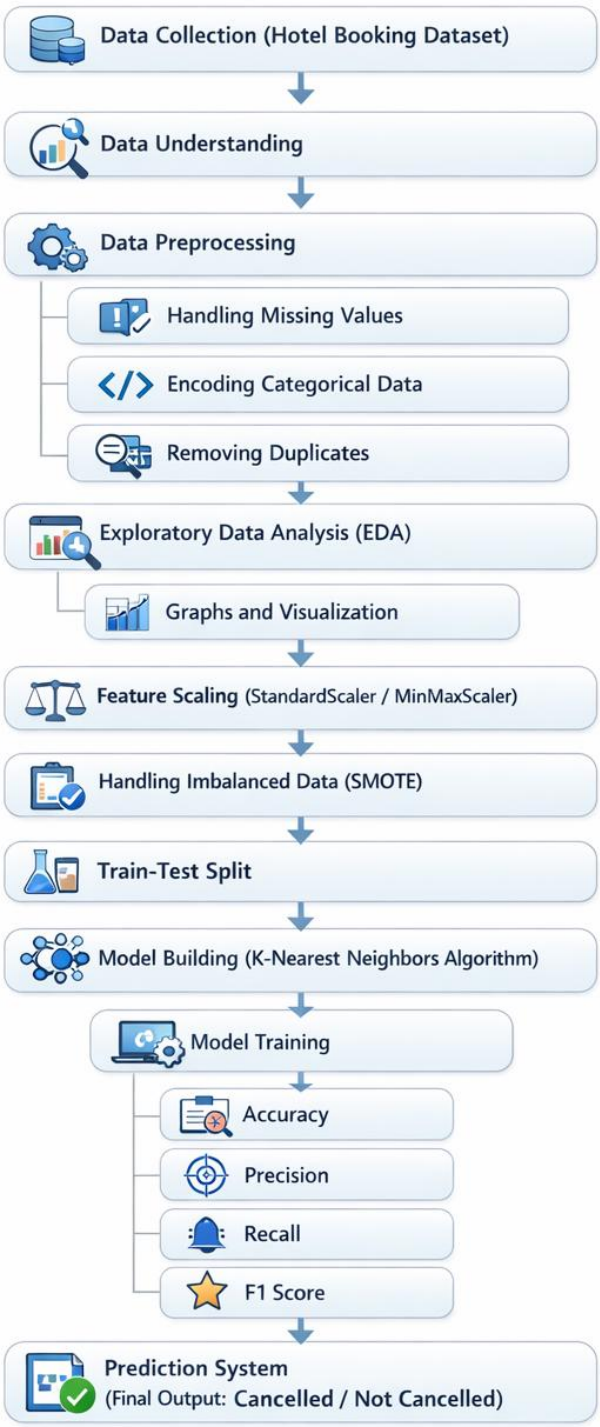
14. Overall Workflow Summary

1. Data Collection
2. Data Cleaning
3. Data Preprocessing
4. EDA
5. Feature Scaling
6. SMOTE
7. Train-Test Split
8. Model Training
9. Model Evaluation
10. Prediction

15. Conclusion of Methodology

The methodology ensures a complete machine learning pipeline, from raw data to prediction. Each step contributes to improving model accuracy, reliability, and practical usability. The use of KNN, combined with preprocessing and balancing techniques, provides a strong foundation for solving the problem of hotel booking cancellations.

Hotel Reservation Cancellation Prediction using KNN Classifier



DATA PREPROCESSING

Data preprocessing is a crucial step in the machine learning pipeline, as the quality of input data directly affects the performance of the model. In this project, raw hotel booking data is cleaned, transformed, and prepared to make it suitable for the K-Nearest Neighbors (KNN) algorithm, which is highly sensitive to data scale and quality.

1. Data Inspection

The first step involves understanding the dataset structure using functions such as:

- .head() → to view initial records
- .info() → to check data types and non-null values
- .describe() → to get statistical summary

This helps identify:

- Missing values
- Data types (numerical/categorical)
- Outliers and inconsistencies

2. Handling Missing Values

Missing values can negatively impact model performance if not handled properly.

Techniques Used:

a) Numerical Features

- Replaced with **mean** or **median**
- Median is preferred when outliers are present

b) Categorical Features

- Replaced with **mode (most frequent value)**

c) Row Removal

- If a row has too many missing values, it is removed

This ensures dataset completeness and consistency.

3. Removing Duplicate Records

Duplicate entries can bias the model by over-representing certain data points.

- Identified using .duplicated()
- Removed using .drop_duplicates()

This step ensures that each booking record is unique.

4. Data Cleaning

This step involves correcting inconsistencies:

- Fixing incorrect data entries
- Standardizing categorical values (e.g., "Online", "online" → same format)
- Ensuring uniform data formatting

5. Encoding Categorical Variables

Machine learning models cannot process categorical data directly, so encoding is required.

5.1 Label Encoding

Used for ordinal categories (having order):

Example:

- Low → 1
- Medium → 2
- High → 3

5.2 One-Hot Encoding

Used for nominal categories (no order):

Example:

- Room Type A → [1,0,0]
- Room Type B → [0,1,0]

This prevents the model from assuming any ranking.

6. Feature Selection

Not all features contribute equally to prediction.

Steps:

- Remove irrelevant columns (e.g., IDs)
- Remove highly correlated features
- Select important features based on:
    - Correlation analysis
    - Domain knowledge

Benefits:

- Reduces overfitting
- Improves model performance
- Decreases computation time

7. Outlier Detection and Treatment

Outliers can distort model predictions.

Detection Methods:

- Box plots
- Z-score method
- IQR (Interquartile Range)

Treatment:

- Remove extreme values
- Cap values within a range

8. Feature Scaling

KNN relies on distance calculation, so scaling is mandatory.

8.1 Why Scaling is Important

Without scaling:

- Features with large values dominate distance calculation

Example:

- Price (1000) vs Number of Guests (2) → price dominates

8.2 Scaling Techniques

a) Standardization (StandardScaler)

- Mean = 0
- Standard deviation = 1

b) Normalization (MinMaxScaler)

- Scales values between 0 and 1

9. Handling Imbalanced Data

The dataset may contain more non-canceled bookings than canceled ones.

9.1 Problem

- Model becomes biased toward majority class
- Poor prediction for minority class

9.2 Solution: SMOTE

SMOTE (Synthetic Minority Oversampling Technique):

- Generates synthetic samples for minority class
- Balances dataset

Steps:

1. Select minority samples
2. Generate new synthetic data
3. Add to dataset

10. Data Splitting Preparation

Before splitting the dataset:

- Features (X) and target (y) are separated
- Preprocessed data is finalized

11. Final Preprocessed Dataset

After preprocessing, the dataset is:

- Clean and consistent
- Fully numerical
- Balanced
- Scaled properly

This dataset is now ready for model training.

12. Importance of Data Preprocessing

Proper preprocessing ensures:

- Higher model accuracy
- Reduced bias and variance
- Better generalization
- Efficient computation

13. Summary

Data preprocessing transforms raw, unstructured data into a clean and meaningful format suitable for machine learning. In this project, it plays a critical role in improving the performance of the KNN classifier by ensuring that all features are properly scaled, encoded, and balanced.

Without effective preprocessing, even the best algorithms fail to produce reliable results, making this step essential for the success of the entire project.

EXPLORATORY DATA ANALYSIS AND VISUALIZATION

EXPLORATORY DATA ANALYSIS (EDA) AND VISUALIZATION

Exploratory Data Analysis (EDA) is an essential stage in the machine learning pipeline that focuses on understanding the structure, distribution, and relationships within the dataset before applying any predictive model. In this project, EDA is performed on the hotel booking dataset to gain meaningful insights into customer behavior and identify the key factors that influence booking cancellations. This step ensures that the data is well understood and that any underlying patterns are properly utilized during model development.

The analysis begins with examining individual features to understand their distribution and characteristics. Numerical variables such as lead time, average price per room, and number of guests are analyzed using statistical summaries and graphical representations like histograms and box plots. These visualizations help in identifying the spread of the data, skewness, and the presence of outliers. For example, lead time often shows a skewed distribution, indicating that a significant number of bookings are made either very early or very close to the arrival date. Similarly, price-related variables may reveal extreme values that need to be handled during preprocessing.

Categorical variables such as room type, meal plan, and market segment are analyzed using frequency-based visualizations like bar charts. This helps in understanding the most common booking preferences and identifying dominant categories within the dataset. For instance, certain room types or booking channels may appear more frequently, indicating customer preferences or business trends. These observations provide valuable insights into how different categories contribute to booking behavior.

Further analysis is conducted to examine the relationship between input features and the target variable, which is the booking cancellation status. This involves comparing how different features behave for canceled and non-canceled bookings. It is observed that bookings with higher lead times tend to have a higher probability of cancellation, as customers making early reservations are more likely to change their plans. Similarly, higher room prices are often associated with increased cancellation rates, suggesting that cost plays a significant role in customer decision-making. On the other hand, repeated guests generally show lower cancellation rates, indicating higher reliability and loyalty.

In addition to individual and pairwise analysis, correlation analysis is performed to understand the relationships among numerical variables. A correlation heatmap is used to visualize how strongly different features are related to each other. This helps in identifying redundant or highly correlated features that may not contribute significantly to the model and

can be removed to improve efficiency. It also highlights important predictors, such as lead time, that have a noticeable relationship with the target variable.

Visualization techniques play a vital role throughout the EDA process by transforming raw data into meaningful graphical representations. These visual tools make it easier to interpret complex data patterns and communicate findings effectively. By using a combination of histograms, box plots, count plots, scatter plots, and heatmaps, the analysis provides a comprehensive understanding of the dataset.

Overall, Exploratory Data Analysis serves as the foundation for building an effective machine learning model. It not only uncovers hidden patterns and trends but also guides the preprocessing and feature selection steps. In this project, EDA significantly contributes to improving the performance of the KNN classifier by ensuring that the model is trained on well-understood and properly structured data. It bridges the gap between raw data and predictive modeling, making it a critical component of the entire workflow.

MODEL BUILDING

Model building is a crucial phase in the machine learning pipeline where the processed data is used to train a predictive model. In this project, the K-Nearest Neighbors (KNN) algorithm is implemented to classify hotel bookings as either canceled or not canceled. The model is developed after completing data preprocessing and exploratory data analysis, ensuring that the input data is clean, properly formatted, and suitable for training.

The process begins by separating the dataset into independent variables (features) and the dependent variable (target). The features include attributes such as lead time, number of guests, room type, pricing, and booking details, while the target variable represents the booking status. Once the data is structured, it is divided into training and testing sets, typically using an 80:20 ratio. The training set is used to build the model, while the testing set is reserved for evaluating its performance on unseen data.

Since the KNN algorithm is based on distance calculations, feature scaling is applied before training the model. Standardization or normalization ensures that all features contribute equally to the distance metric, preventing variables with larger numerical ranges from dominating the model's predictions. This step is essential for achieving accurate and reliable results.

The K-Nearest Neighbors algorithm works by storing all training data points and making predictions based on the similarity between new input data and existing data points. When a new booking record is provided, the algorithm calculates the distance between this record and all data points in the training set, typically using the Euclidean distance metric. It then selects the 'K' nearest neighbors and assigns the class based on majority voting among those neighbors.

Selecting the optimal value of K is an important part of model building. A small value of K can make the model sensitive to noise and lead to overfitting, while a large value can smooth out important patterns and result in underfitting. Therefore, multiple values of K are tested, and the one that provides the best performance on validation data is chosen. This process may involve comparing accuracy scores or using cross-validation techniques.

To further improve model performance, techniques such as handling class imbalance using SMOTE are incorporated before training. This ensures that the model does not become biased toward the majority class and can accurately predict both canceled and non-canceled bookings.

Once the optimal parameters are selected, the final KNN model is trained using the processed training dataset. The trained model is then used to make predictions on the test dataset. The predictions are compared with actual values to evaluate the effectiveness of the model.

Overall, the model building process in this project focuses on constructing a robust and reliable KNN classifier by carefully preparing the data, selecting appropriate parameters, and ensuring proper evaluation. This structured approach helps in achieving accurate predictions and

demonstrates the practical application of machine learning in solving real-world problems in the hospitality industry.

MODEL EVALUATION

Model evaluation is a critical phase in the machine learning pipeline that assesses the performance and reliability of the trained model. In this project, the K-Nearest Neighbors (KNN) classifier is evaluated using the test dataset to determine how effectively it can predict hotel booking cancellations on unseen data. This step ensures that the model is not only accurate but also generalizes well beyond the training data.

After training the model, predictions are made on the test dataset, and these predicted values are compared with the actual booking status. The evaluation process begins with the construction of a confusion matrix, which provides a detailed breakdown of correct and incorrect predictions. It categorizes the results into true positives, true negatives, false positives, and false negatives. This matrix helps in understanding how well the model distinguishes between canceled and non-canceled bookings and identifies the types of errors being made.

To quantify the model's performance, several evaluation metrics are used. Accuracy measures the overall correctness of the model by calculating the proportion of correctly predicted instances out of the total predictions. While accuracy provides a general idea of performance, it may not be sufficient when dealing with imbalanced datasets. Therefore, additional metrics such as precision and recall are considered. Precision indicates how many of the predicted cancellations were actually correct, while recall measures how effectively the model identifies all actual cancellations. These metrics are particularly important in this project, as missing a cancellation (false negative) can have a significant impact on hotel revenue.

The F1-score is also used as a balanced metric that combines both precision and recall. It provides a single measure that reflects the trade-off between these two metrics, making it especially useful when the dataset is imbalanced. A higher F1-score indicates better model performance in terms of both identifying cancellations and minimizing false predictions.

In addition to these metrics, the effect of different values of K in the KNN algorithm is analyzed during evaluation. By testing multiple values of K, the model's performance is compared, and the optimal value is selected based on the best balance between bias and variance. This helps in improving the overall accuracy and robustness of the model.

The evaluation results show that proper preprocessing, feature scaling, and handling of class imbalance significantly improve the performance of the KNN classifier. The model demonstrates a reasonable ability to predict booking cancellations, with improved accuracy and balanced performance across different evaluation metrics.

Overall, model evaluation provides a comprehensive understanding of the strengths and limitations of the KNN model. It ensures that the model is reliable and suitable for practical use, while also highlighting areas for further improvement. This step is essential for

validating the effectiveness of the machine learning approach in solving the problem of hotel booking cancellation prediction.

```
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Prediction
y_pred = model.predict(X_test)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:\n", cm)

# Classification Report
print("\nClassification Report:\n", classification_report(y_test, y_pred))
```

FINAL PREDICTION SYSTEM

The final prediction system represents the practical implementation of the trained machine learning model, where the K-Nearest Neighbors (KNN) classifier is used to predict whether a new hotel booking will be canceled or not. This system integrates all the stages of the machine learning pipeline, including preprocessing, feature transformation, and model inference, into a unified workflow that can be used for real-world predictions.

The system begins by accepting new booking data as input. This input includes all relevant features such as the number of guests, lead time, room type, pricing details, market segment, and any special requests. Since the model was trained on preprocessed data, the same preprocessing steps must be applied to the new input data to maintain consistency. This includes handling any missing values, encoding categorical variables into numerical format, and applying feature scaling using the same scaler (such as StandardScaler) that was used during model training.

Once the input data is properly preprocessed, it is passed to the trained KNN model. The model then calculates the distance between the new booking instance and all data points in the training dataset. Based on the selected value of K, it identifies the nearest neighbors and performs majority voting to determine the predicted class. The output of the model is a binary result indicating whether the booking is likely to be canceled or not canceled.

The prediction system is designed to be efficient and user-friendly, allowing hotel management or booking platforms to quickly assess the risk associated with each reservation. This information can be used to make strategic decisions, such as applying dynamic pricing, managing overbooking, or offering incentives to reduce the likelihood of cancellation.

In addition, the system can be extended to provide probability-based outputs, indicating the likelihood of cancellation rather than just a binary result. This allows for more nuanced decision-making and better risk assessment. The model can also be integrated into a web-based application using frameworks such as Flask or Streamlit, enabling real-time predictions through a user interface.

Overall, the final prediction system transforms the developed machine learning model into a practical tool that can be used in real-world scenarios. By accurately predicting booking cancellations, it helps improve operational efficiency, optimize revenue management, and enhance customer satisfaction in the hospitality industry.

CONCLUSION

This project successfully demonstrates the application of machine learning techniques in predicting hotel booking cancellations using the K-Nearest Neighbors (KNN) algorithm. By utilizing historical booking data and following a structured data science pipeline, the system is able to identify patterns and factors that influence customer cancellation behavior.

Throughout the project, various stages such as data preprocessing, exploratory data analysis, feature scaling, and handling of imbalanced data were carefully implemented to improve the quality of the dataset and enhance model performance. The use of EDA provided valuable insights into important features like lead time, pricing, and customer type, which significantly impact cancellation trends.

The KNN model, being simple and intuitive, proved effective in handling this classification problem when combined with proper preprocessing and parameter tuning. The evaluation metrics, including accuracy, precision, recall, and F1-score, indicate that the model performs reasonably well in predicting booking outcomes. Additionally, techniques like SMOTE contributed to improving the model's ability to handle imbalanced data and make fair predictions for both classes.

The final prediction system developed in this project demonstrates practical applicability, as it can assist hotel management in making informed decisions related to pricing strategies, overbooking, and resource allocation. By predicting potential cancellations in advance, hotels can minimize revenue loss and improve operational efficiency.

In conclusion, this project highlights the importance of data-driven decision-making in the hospitality industry and showcases how machine learning can be effectively used to solve real-world business problems. While the KNN model provides satisfactory results, there is scope for further improvement using advanced algorithms and optimization techniques. Overall, the project serves as a strong foundation for developing intelligent predictive systems in the domain of hotel management.

FUTURE SCOPE

The current project demonstrates the use of the K-Nearest Neighbors (KNN) algorithm for predicting hotel booking cancellations. However, the system can be further improved and extended in several ways to enhance its performance and real-world applicability.

Firstly, more advanced machine learning algorithms such as **Logistic Regression, Decision Trees, Random Forest, and XGBoost** can be implemented and compared with KNN to achieve better accuracy and performance. Hyperparameter tuning techniques can also be applied to optimize the model.

Secondly, the project can be enhanced by using a larger and more detailed dataset with additional features such as customer reviews, seasonal trends, and booking sources. This can improve the predictive capability of the model and provide deeper insights into customer behavior.

The user interface can also be improved by developing a more advanced web application with better design and user experience. The system can be deployed online so that it can be accessed by users in real-time.

Additionally, the project can be integrated with real hotel booking systems to provide live predictions and assist in decision-making processes such as pricing strategies and overbooking management.

Overall, the project has strong potential to be developed into a complete intelligent hotel management system using advanced analytics and real-time data.

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